

Kubernetes Volume Types

A Kubernetes **volume** provides storage or configuration data that can be used by containers inside a Pod.

Basic Syntax

```
volumes:
  - name: <volume-name>
    <volume-type>:
      <configuration>
```

Mounting the Volume into a Container

```
volumeMounts:
  - name: <volume-name>
    mountPath: <path-inside-container>
```

1. ConfigMap

Used to provide **configuration files or values** to a Pod.

```
volumes:
  - name: config-volume
    configMap:
      name: app-config
```

Example mount:

```
volumeMounts:
  - name: config-volume
    mountPath: /etc/config
```

Use: Application configuration.

2. Secret

Used to provide **sensitive configuration**, such as passwords, tokens, or certificates.

```
volumes:
  - name: secret-volume
    secret:
      secretName: app-secret
```

Example mount:

```
volumeMounts:
  - name: secret-volume
    mountPath: /etc/secrets
```

Use: Sensitive data.

3. emptyDir

Creates an **empty temporary directory** when the Pod starts.

```
volumes:
  - name: temp-storage
    emptyDir: {}
```

Example mount:

```
volumeMounts:
  - name: temp-storage
    mountPath: /tmp/app
```

Use: Temporary files, caching, scratch space.

Note: Data is deleted when the Pod is removed.

4. PersistentVolumeClaim (PVC)

Used when the application needs **persistent storage**.

```
volumes:
  - name: app-storage
    persistentVolumeClaim:
      claimName: app-pvc
```

Example mount:

```
volumeMounts:
  - name: app-storage
    mountPath: /data
```

Use: Databases, persistent application data, and files that should survive Pod restarts.

5. hostPath

Mounts a directory or file from the **Kubernetes node** into the Pod.

```
volumes:
  - name: host-storage
    hostPath:
      path: /var/log/myapp
      type: DirectoryOrCreate
```

Use: Special node-level workloads, monitoring agents, etc.

Note: Generally avoid it for normal application workloads because the data is tied to a particular node.

Quick Comparison

Volume type	Main purpose	Data survives Pod deletion?
configMap	Configuration	ConfigMap survives
secret	Sensitive configuration	Secret survives
emptyDir	Temporary storage	No
persistentVolumeClaim	Persistent application data	Usually
hostPath	Node filesystem access	Depends on the node

Easy Way to Remember

```
ConfigMap    → Configuration
Secret       → Sensitive data
emptyDir     → Temporary data
PVC          → Persistent data
hostPath     → Node's filesystem
```

Important: A volume is defined under the Pod's volumes: section, but it becomes available to a specific container only when you add the corresponding volumeMounts: entry.